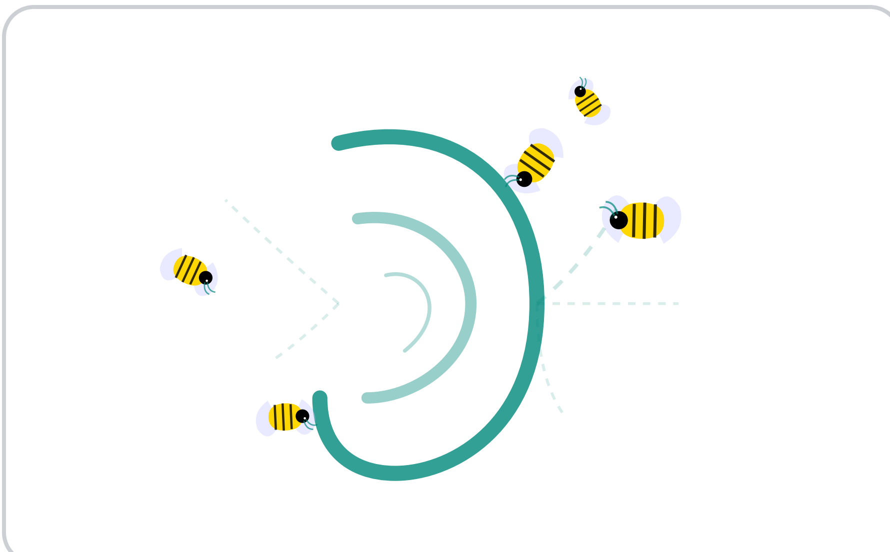
# TINNITUS THERAPY SUITE

## User Manual



Version 2026.05.4

Open-source, browser-based sound therapy tools for tinnitus research and experimentation.

Tinnitus Therapy Suite | User Manual Generated: 5/30/2026

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# SAFETY & LEGAL DISCLAIMER



DISCLAIMER

Official Web Access: tinnitus.trahreg.com

This tinnitus therapy suite is an experimental, educational, and informational tool.
It is not a medical device and is not intended to diagnose, treat, cure, or prevent
any disease or health condition.

No medical claims are made or implied. The sound tools provided here are based on
general concepts found in tinnitus research, but they are not a substitute for
professional evaluation or treatment by a licensed clinician, audiologist, or
healthcare provider.

AI-GENERATED CONTENT:
Certain features (CBT Reframing, SOS Spike Support, Sound Recipes, Pattern Identification, and Clinical Summaries) utilize Artificial Intelligence (Google Gemini). AI can "hallucinate" or provide inaccurate information. All AI-generated content is for educational exploration only. Do not make medical decisions based on AI suggestions without consulting a professional.

Use of this software is voluntary and at your own risk. Always keep listening volume
at a safe and comfortable level. Stop using the tool immediately if you experience
pain, discomfort, dizziness, or worsening symptoms.

If you have concerns about tinnitus, hearing loss, hyperacusis, or any other medical
condition, consult a qualified healthcare professional.

# SAFETY & LEGAL DISCLAIMER



```
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The authors and contributors are not liable for any damages, losses, or
adverse
effects arising from the use of this software.

By using this tool, you acknowledge that you have read and understood
this disclaimer.
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*Note: Acceptance of this disclaimer is required before utilizing any therapeutic modules within the suite.*

# 1. INTRODUCTION & SUCCESS PROTOCOLS

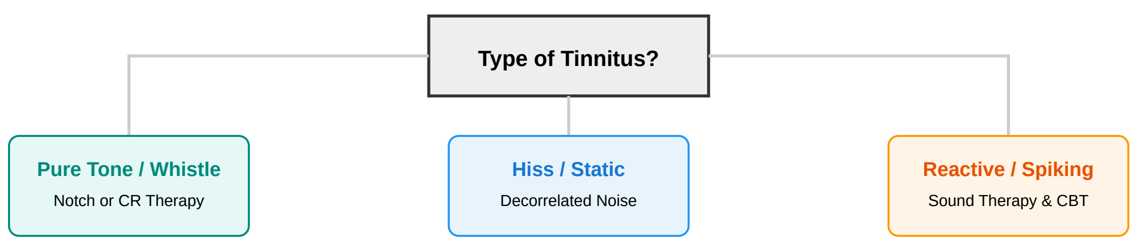
The Trahreg Tinnitus Therapy Suite is a research-oriented platform built on three core pillars:

- **Evidence-Based Tools:** Digital implementations of established protocols like Notch Therapy and Acoustic CR.
- **Clinical Transparency:** Tools to measure Minimum Masking Levels (MML) and track THI progress.
- **Privacy-First Design:** A "Zero-Knowledge" architecture where all data stays on your local device.

Our goal is to facilitate **Habituation**—the neuroplastic process where the brain learns to reclassify the tinnitus signal as a neutral, non-threatening background sound.

## Therapy Selection Matrix

FLOWCHART: SELECTING A MODALITY




## Before You Begin: System Readiness

☐ **Hardware:** Use high-quality, **wired, over-ear headphones**. Open-back models are preferred for natural spatial imaging. Avoid Bluetooth "Hands-Free" (AG) mode.

☐ **OS Audio:** Disable "Spatial Sound," "Dolby Atmos," "Windows Sonic," or any "Bass Boost" effects.

# 1. INTRODUCTION & SUCCESS PROTOCOLS

☐ **Sample Rate:** Ensure your system is set to **44.1 kHz** or **48 kHz**. High-frequency filters require this precision to remain stable.

☐ **Safe Levels:** Sound therapy should never be painful. Aim for a listening level **below 70 dB** (comfortable conversation level).

☐ **Environment:** Conduct your calibration in a quiet room to ensure accurate **Pitch Matching**.

## Which Therapy Should I Use? (Selection Matrix)

| If your symptoms are... | Recommended Starting Point |
|---|---|
| **High Distress / Anxiety / "Spikes"** | **CBT & Wellness** + 4-7-8 Breathing Pacer |
| **A constant, high-pitched "Whistle" or "Tone"** | **Notch Therapy** or **Acoustic CR** |
| **A broad "Hiss", "Static", or "Steam" sound** | **Decorrelated Noise** or **Broadband Sound** |
| **Reactive (gets louder with environmental noise)** | **Broadband Sound** (Set to very low volume) |
| **Significant difficulty falling asleep** | **Binaural Beats** (Delta/Theta settings) |

## The Five Golden Rules

- **Rule 1: The Mixing Point.** Therapy volume must be set so the sound and your tinnitus "mix" like two colors of paint. Total masking (hiding the tinnitus completely) prevents habituation; your brain must still perceive the signal to learn it is neutral.

VISUAL: THE MIXING POINT

# 1. INTRODUCTION & SUCCESS PROTOCOLS

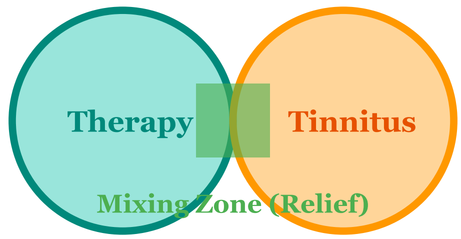



*Figure: Both signals must be audible for the brain to habituate.*

- **Rule 2: Passive Listening.** Treat therapy like background "wallpaper." Do not focus on the sound; instead, engage in work, reading, or relaxation while the engine runs.
- **Rule 3: Consistency.** Neuroplasticity takes time. Aim for 30–60 minutes of focused therapy daily for at least 3 months.
- **Rule 4: AI Support.** If you experience a "spike" in distress, use the AI-powered **CBT Assistant** for immediate reframing and coping strategies rather than simply increasing the volume.
- **Rule 5: Clinical Visibility.** Regularly export your **Clinical PDF Reports** to share objective progress (THI and MML trends) with your audiologist.
- **Rule 6: Audio Integrity.** You must disable OS-level enhancements like Windows "Sonic," Dolby Atmos, or "Bass Boost," as these distort the calibrated therapy filters.

## Protocol for Success (Phases)

1. **Phase 1: Validation.** Run **System Validation** weekly. This ensures your headphones are in-phase and the browser audio engine is delivering accurate signals.
2. **Phase 2: Pitch Match.** Use the **Notch Finder**. Accuracy is critical; if your **Pitch Match** is off by more than 5%, Notch and CR therapies lose clinical effectiveness.
3. **Phase 3: Calibration.** Set your therapy volume to the **Mixing Point** (slightly *lower* than your tinnitus). Total masking prevents the brain from habituating.
4. **Phase 4: Active Habituation.** Use the session timer and engage in non-auditory tasks. The goal is for the sound to fade into the background.
5. **Phase 5: Data Correlation.** Review "AI Insights" weekly to identify if sleep, stress, or environment are triggering spikes.

# 1. INTRODUCTION & SUCCESS PROTOCOLS

6. **Phase 6: Monthly Review.** Every 30 days, generate a Clinical PDF. Look for a 7-point drop in THI scores as a marker of success.

## Using the Help Systems

- **The Guide:** Accessible via the "Clinical Guide" button, this panel provides immediate access to the Golden Rules and Success Protocols from any page.
- **Interactive Tutorials:** Click the "Tutorial" button on any therapy page for a step-by-step narrated walkthrough.
- **Professional Narrator:** High-quality audio guidance is available for all setup steps. You can adjust the narrator's speed and volume in the tutorial menu.
- **AI Assistant:** Integrated into the CBT and Insight modules for real-time reframing and sound recipe suggestions.

## 2. DECORRELATED NOISE THERAPY

**Purpose:** Tinnitus is often associated with pathological neural synchrony. Decorrelation presents independent, unrelated noise signals to each ear, preventing the brain from "merging" the sound into a central image. This aims to disrupt the synchronized firing patterns in the auditory cortex.

**QUICK SETUP**

- **Hardware:** High-quality headphones required.
- **Action:** Adjust **Decorrelation** slider until the sound feels "wide" or "filling the head."

### Step-by-Step Tutorial

1. **Hardware:** Connect high-quality headphones. True decorrelation requires two discrete signals.
2. **Sound Choice:** Select a noise color (**Pink** is recommended for long-term comfort).
3. **Calibration:** Adjust the **Decorrelation** slider until the sound feels "wide" and fills your head, rather than being a single point.
4. **Volume:** Set to the **Mixing Point**.

> ***Clinical Example:*** *"My tinnitus sounds like broad steam or static in both ears."*
> ***Setup:*** *Select* ***Pink Noise****, set* ***Decorrelation*** *to 80% for a 'wide' soundstage, and adjust volume to the* ***Mixing Point****.*

### Field Guide

- **Decorrelation Slider:** Controls the independence of the L/R channels. 0% is identical (Mono); 100% is fully unrelated noise.
- **Noise Color:**
  - **Pink:** Equal energy per octave; mirrors natural sound distribution.
  - **White:** Equal energy per frequency; effective for aggressive masking.
  - **Brown:** Heavy low-frequency emphasis; best for low-pitched "roaring" tinnitus.

## 2. DECORRELATED NOISE THERAPY

- **EQ Sliders:** Manually boost frequencies (2kHz to 12kHz) where you have known hearing loss to reduce "listening effort."

### Pro-Tips

- If decorrelation feels "dizzying," reduce the slider to 50% until your brain adjusts to the wider soundstage.

# 3. NOTCH THERAPY

**Purpose:** Based on **Tailored Notched Music Training (TMNMT)**, this therapy removes sound energy from a narrow band centered on your tinnitus pitch. This stimulates **Lateral Inhibition**: the stimulated neurons *surrounding* the notch actively suppress the hyperactive, tinnitus-producing neurons *inside* the notch. *(Ref: Okamoto et al., 2010)*

**QUICK SETUP**

- **Calibration:** Match your pitch exactly in the **Notch Finder**.
- **Action:** Set **Notch Width** to 1.0 octaves and find the **Mixing Point**.

## Step-by-Step Tutorial

1. **Pitch Match:** Use the **Notch Finder** to determine your **Center Frequency**.
2. **Frequency Entry:** Input that frequency into the **Notch Center** field.
3. **Width:** Ensure **Notch Width** is set to **1.0 octaves** (the clinical standard).
4. **Mixing:** Set the volume so you can still perceive your tinnitus "inside" the silent notch in the noise.

> ***Clinical Example:*** *"My tinnitus is a sharp 8,400 Hz whistle."*
> ***Setup:*** *Use* ***Notch Finder*** *to verify the 8,400 Hz* ***Pitch Match****. Set* ***Notch Center*** *to 8,400 Hz and* ***Width*** *to 1.0 octaves.*

## Field Guide

- **Notch Center (Hz):** The frequency of your tinnitus. Every slider movement here shifts the "silence" in the sound.
- **Notch Width (Octaves):** The size of the frequency "hole." 1.0 is standard; smaller widths require higher pitch-matching precision.
- **Source Selector:** Choose between **Internal Noise**, a **Custom MP3**, or **Live Audio** (e.g., Spotify processed through the notch).

# 3. NOTCH THERAPY

## Pro-Tips

- Re-verify your **Pitch Match** monthly. Tinnitus pitch can shift slightly as habituation begins.

## 4. DUAL-STIMULUS NEUROMODULATION

**Purpose:** Inspired by bimodal devices like **Lenire**, this experimental module pairs structured auditory patterns (carrier tones, modulated noise, and tone bursts) with tactile feedback (haptics). By stimulating both the auditory and somatosensory systems simultaneously, the brain is encouraged to re-prioritize external signals over the internal tinnitus signal. *(Ref: TENT-A Trial, 2020)*

**QUICK SETUP**

- **Hardware:** Use on a smartphone held in hand for tactile feedback.
- **Action:** Enable **Tactile Haptic Pulse** and sync the **Pulse Rate** to your breathing.

### Step-by-Step Tutorial

1. **Device:** Best used on a smartphone held in the hand.
2. **Activation:** Enable **Tactile Haptic Pulse**.
3. **Sensitivity:** Adjust the **Trigger Sensitivity** until the phone vibrates *only* when you hear a distinct audio burst.
4. **Rhythm:** Use the **Pulse Rate** slider to match the visual pacer to your resting breathing rate (usually 5-6 BPM).

> ***Clinical Example:*** *"Sound therapy alone isn't enough to distract me from my tinnitus."*
> ***Setup:*** *Enable **Tactile Haptic Pulse**. Set **Pulse Rate** to 6 BPM and hold the device to engage the somatosensory system.*

### Field Guide

- **Carrier Freq (Hz):** Sets the pitch of the tone bursts. Ideally set near your tinnitus frequency or a complementary low tone.
- **Noise Mod (Hz):** The speed at which the background noise "shimmers" (amplitude modulation).

## 4. DUAL-STIMULUS NEUROMODULATION

- **Burst Rate (Hz):** How many audio pulses occur per second.
- **Tone/Noise Mix:** Slides between pure bursts (1.0) and pure modulated noise (0.0). A 50/50 mix is standard.
- **Haptic Strength:** Controls how forcefully the phone vibrates during a trigger.

### Pro-Tips

- Use a phone case with good vibration transmission to maximize the somatosensory effect.

# 5. BROADBAND SOUND THERAPY

**Purpose:** The foundational tool for **Tinnitus Retraining Therapy (TRT)**. It provides constant, neutral sound enrichment to reduce the "contrast" between the tinnitus signal and a silent environment, preventing the brain's internal gain from increasing.

**QUICK SETUP**

- **Choice:** Select **Pink Noise** or **Rain**.
- **Action:** Use the **Mixing Point** rule for passive, long-term enrichment.

## Step-by-Step Tutorial

1. **Selection:** Choose a sound you find neutral (**Pink Noise**) or pleasant (**Rain/Ocean**).
2. **Calibration:** Use the **Mixing Point** rule. The sound should be audible but *not* hide the tinnitus.
3. **Duration:** Use for 24/7 enrichment at low levels, or set a 60-minute timer for a focused session.

> ***Clinical Example:*** *"I have reactive tinnitus that gets louder in silent rooms."*
> ***Setup:*** *Select* ***Rain*** *or* ***Pink Noise****. Set volume very low (just above audibility) for constant environment enrichment.*

## Field Guide

- **Noise Selection:** Buttons for White, Pink, Brown, Blue, and Violet. Pink is the research standard for habituation.
- **L/R Balance:** Moves the sound toward your "tinnitus side." Important if your tinnitus is louder in one ear.
- **Sleep Fade:** When checked, the volume will slowly ramp down to zero over 60 seconds at the end of your timer to prevent an autonomic "rebound" spike.
- **4-7-8 Pacer:** A visual and auditory breathing guide. Inhale for 4s, Hold for 7s, Exhale for 8s. Use during high-distress spikes.

# 5. BROADBAND SOUND THERAPY

## Pro-Tips

- If **Pink Noise** feels too sharp, switch to **Brown Noise**, which has a smoother, deeper profile similar to heavy rain.

# 6. ACOUSTIC COORDINATED RESET (CR)

**Purpose:** Developed by **Prof. Peter Tass**, this neuromodulation protocol uses pseudo-random sequences of four tones mathematically calculated relative to your tinnitus pitch. The goal is to desynchronize the pathological neural loops responsible for tonal tinnitus. *(Ref: Tass et al., 2012)*

**DIAGRAM: CR TONE DISTRIBUTION**

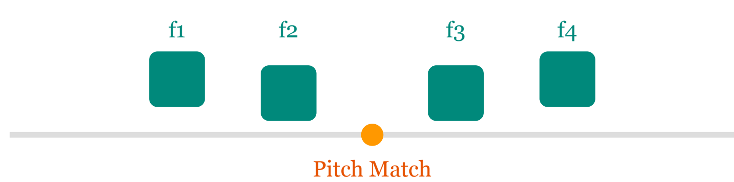



*Figure: Four tones (f1-f4) are mathematically arranged around your tinnitus pitch.*

### QUICK SETUP

- **Action:** Click **Calculate** after pitch matching and adjust **Bal 1-4** for equal loudness.
- **Protocol:** Follow the 3-on/2-off sequence for 60 minutes daily.

## Step-by-Step Tutorial

1. **Pitch Match:** Use the **Pitch Bracketing** tool to find your exact **Center Frequency**.
2. **Calculation:** Enter this frequency into the **Center Frequency** field and click "Calculate."
3. **Balancing:** This is critical. Play the sequence and adjust the **Bal 1-4** sliders until all four tones sound *exactly as loud as each other* to your ears.
4. **Protocol:** Follow the "3-on/2-off" pattern. 60-minute sessions are recommended.

> ***Clinical Example:*** *"I have a single, constant tone at 4,000 Hz."*
> ***Setup:*** *Enter 4,000 Hz as the* ***Base Frequency****. Adjust* ***Bal 1-4*** *until all four therapeutic tones sound equally loud.*

# 6. ACOUSTIC COORDINATED RESET (CR)

## Field Guide

- **Tone Grid:** Displays the active therapeutic frequencies. A highlighted box indicates which tone is currently firing.
- **Bal Sliders (1-4):** Adjusts individual tone volumes. Use these to compensate for hearing loss at specific frequencies.
- **Cycle Dots:** A row of 5 dots. 3 solid dots (Active cycles) and 2 dashed dots (Silence cycles). This ensures the brain does not "adapt" to the sequence.

### Pro-Tips

- Do not use CR tones as a masking sound. They should be played at a comfortable, audible level, but generally lower than the **Mixing Point** used for noise.

# 7. BINAURAL BEATS & WELLNESS

**Purpose:** Utilizes the **Frequency Following Response (FFR)**. By playing two slightly different frequencies in each ear, the brain "perceives" a third, internal beat frequency. This can be used to induce Alpha, Theta (relaxation), or Delta (deep sleep) states, helping to down-regulate a stressed nervous system.

**QUICK SETUP**

- **Hardware:** Stereo headphones are strictly required.
- **Target:** Choose **Delta** for sleep or **Theta/Alpha** for relaxation.

## Step-by-Step Tutorial

1. **Hardware:** Ensure stereo headphones are connected.
2. **Selection:** Choose **Delta** for sleep or **Theta** for relaxation.
3. **Calibration:** Set volume to a level that is comfortable and soothing.
4. **Listening:** Close your eyes and focus on your breathing.

> ***Clinical Example:*** *"Tinnitus makes it very hard to fall asleep."*
> ***Setup:*** *Choose the* ***Delta*** *preset. Use over-ear headphones and set a 30-minute timer with* ***Sleep Fade*** *enabled.*

## Field Guide

- **Stereo Separation:** You MUST use headphones. Without discrete L/R separation, the binaural effect is lost.
- **Delta (1-4Hz):** Induces deep, restorative sleep.
- **Theta (4-8Hz):** Induces meditative, relaxed states; ideal for anxiety management.
- **Alpha (8-13Hz):** Promotes calm focus.

## 7. BINAURAL BEATS & WELLNESS

### Pro-Tips

- Binaural beats can be layered with **Broadband Sound** for enhanced comfort during sleep transitions.

## 8. TINNITUS HANDICAP INVENTORY (THI)

**Purpose:** A 25-item clinically validated tool that measures the functional, emotional, and catastrophic impact of tinnitus on your life. It categorizes severity into Grade 1 (Slight) through Grade 5 (Catastrophic).
**Usage:** Take the assessment every 30 days. Your progress is measured by a reduction in this score, not necessarily a change in the volume of the sound.

## 9. COGNITIVE RESTRUCTURING

**Purpose:** Tinnitus distress is often driven by the **Limbic System** (the brain's emotion center). This tool uses interactive **Thought Records** to help you identify negative "Automatic Thoughts" and reframe them into objective "Balanced Thoughts," facilitating habituation.

### How to Use

1. **Log:** Record a situation where tinnitus was distressing.
2. **Identify:** Write down the thought you had (e.g., "This will never stop").
3. **AI Assist:** Use the **Balanced Thought** button to generate a clinical TRT reframing suggestion.

## 10. RELAXATION TRAINING

**Purpose:** Tinnitus and physiological stress exist in a feedback loop. This module provides **Progressive Muscle Relaxation (PMR)** guides and the **4-7-8 Breathing Pacer** to de-escalate the autonomic nervous system.

## 11. PERSONALIZED RECOMMENDATIONS

**Purpose:** An algorithmic tool that analyzes your THI scores and MML levels to suggest the most effective therapy for your specific grade of tinnitus. For example, high distress levels will prioritize CBT and Relaxation, while lower distress with tonal tinnitus will prioritize Notch therapy.

## 11. PERSONALIZED RECOMMENDATIONS

### Mental Health Spike Protocol

If a tinnitus "spike" causes high emotional distress, clinicians recommend switching focus from Sound Therapy to **CBT & Wellness**. Managing the limbic system's reaction—getting your body out of "fight or flight"—is more important than the acoustic signal during a spike.

# 12. NOTCH FINDER

**Purpose:** Accurate pitch matching is the foundation of Notch and CR therapies. The **Notch Finder** identifies your "Center Frequency"—the specific pitch of your tinnitus.

DIAGRAM: NOTCH FINDER CALIBRATION

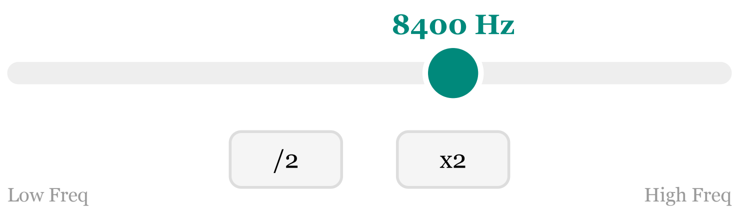



*Figure: Use the slider to find a match, then use /2 and x2 to verify the fundamental pitch.*

## How to Match Your Pitch

1. **Environment:** Sit in a quiet room with headphones.
2. **Search:** Move the **Frequency Slider** slowly until the external tone matches your internal sound to find the **Pitch Match**.
3. **Bracketing Strategy:** Once you find a potential match, move the slider 500Hz above and below. If the tinnitus "disappears" or "beats" with the tone, you are close.
4. **Octave Verification:** This is the most critical step. Click **/2 (Half)**. If the lower tone still sounds like your tinnitus, the original was a harmonic. The **lowest** clear match is your fundamental frequency.
5. **Two-Tone Comparison:** If you are stuck, use the **Two-Tone** module to compare Tone A and Tone B to "bracket" the sound.

**TECHNICAL NOTE: OCTAVE ERRORS**

## 12. NOTCH FINDER

Matching your pitch to an octave higher than the actual signal is common. If your Notch therapy feels "piercing" rather than "soothing," re-run the **/2** check. Clinical efficacy drops by 70% if the notch is set to the wrong octave.

### Field Guide

- **Frequency (Hz):** The current test pitch. Can be entered manually for precision.
- **Auto-Sweep Speed:** Sets the rate at which the tone automatically climbs the frequency range. Many users find it easier to identify their pitch while the sound is moving.
- **Save Frequency:** Saves the match globally so it is automatically applied to Notch and CR therapy modules.

## 13. TWO-TONE COMPARISON

**Purpose:** Diagnostic aid to avoid octave errors. Allows A/B comparison between two distinct frequencies.

## 14. FREQUENCY SWEEP

**Purpose:** A 10-second logarithmic sweep from 20Hz to 20,000Hz. Useful for identifying "dead zones" in hearing or specific frequencies that trigger a "reactive" increase in tinnitus volume.

## 15. TINNITUS MASKING CURVE (TMC)

Maps your **Minimum Masking Level (MML)** across the spectrum. A "steep" curve calculates a high **Q-factor**, indicating tonal tinnitus, while a flat curve suggests broadband noise-like tinnitus.

## 16. LOUDNESS GROWTH (LG)

**Purpose:** Measures how your subjective perception of loudness grows relative to objective volume increments. This detects **Hyperacusis** (sound sensitivity) or **Recruitment**. *Tutorial: Play the tone and rate its loudness from "Very Soft" to "Uncomfortable."*

## 17. SUPPRESSION TEST (RI)

**Residual Inhibition (RI):** Measures the temporary silence or volume reduction after 60 seconds of high-level sound exposure.

1. **Step 1:** Play noise at a level that completely hides your tinnitus for 60 seconds.
2. **Step 2:** The noise stops. Use the stopwatch to time how many seconds it takes for your tinnitus to return to normal.
3. **Insight:** Increasing RI duration over months is a positive indicator of auditory system responsiveness.

## 18. HEARING RANGE & AUDIOGRAM PROFILE

**Hearing Range Test:** Tinnitus is often a "compensation" for auditory deprivation. Use this tool to explore your audibility thresholds across the spectrum (20Hz to 20kHz) and identify shifts in hearing.

# 18. HEARING RANGE & AUDIOGRAM PROFILE

## Hearing Profile (Audiogram)

The suite allows you to enter your professional audiogram results (0dB to 110dB HL) for frequencies between 250Hz and 12kHz. This data is used to personalize your therapy.

- **The Half-Gain Rule:** A clinical standard for hearing compensation. The suite calculates a therapeutic boost equal to half of your recorded hearing loss (e.g., a 40dB loss results in a 20dB boost).
- **Digital Safety Ceiling:** To prevent clipping or acoustic trauma, the maximum digital boost is capped at **20dB**.
- **Auto-Compensation:** Once saved, your profile automatically adjusts the output of the **Sound Therapy**, **Notch**, and **Lenire-style** modules to ensure frequencies you struggle to hear are properly stimulated.

**CLINICAL RECOMMENDATION**

Update your digital Hearing Profile every 6-12 months or following a professional hearing test. Properly compensating for hearing loss reduces "listening effort" and can significantly lower the brain's internal gain, aiding habituation.

## Tinnitus Side Preset

On the Audiogram page, you can select your **Tinnitus Side** (Left, Right, or Both). This sets a global intelligent default for the L/R balance in all therapy modules, ensuring the sound is immediately centered or biased toward the appropriate ear upon starting a session.

## 19. AI-POWERED FEATURES

**AI-Powered Features (Gemini):** Leverages client-side AI (via your own API key) to provide therapeutic support. All data is anonymized before processing.

- **"Spike" De-escalator:** Provides immediate CBT reframing during periods of high distress.
- **Soundscape Designer:** Suggests "Sound Recipes" based on natural language descriptions (e.g., "high-pitched hiss" -> "Pink Noise with Stream").
- **Pattern Correlation:** Identifies non-obvious links between your sleep, usage, and distress scores.
- **Clinical Report Translator:** Generates a professional summary of your progress for your doctor.
- **Grounding Script Generation:** Located in the Thought Record module, this tool generates a personalized 1-minute mindfulness script based on your specific "Balanced Thought."

## 20. SPECTROGRAM & LEVEL METER

**Spectrogram:** A real-time visualizer of audio frequencies. Use this to verify your **Notch** (look for the "hole" in the graph) or **Acoustic CR** tone alignment.
**Audio Level Meter:** Displays real-time environmental volume. Habituation is most effective in environments between 45dB and 60dB.

## 21. FEEDBACK TOOL

The feedback tool allows users to submit technical bug reports or feature requests. To maintain clinical privacy, you can generate a technical log file to send via email rather than using a persistent database or third-party tracking.

## 22. SYSTEM VALIDATION ENGINE

High-frequency therapy requires extreme technical precision. This engine audits your hardware and browser settings weekly:

**DIAGRAM: VALIDATION DASHBOARD**

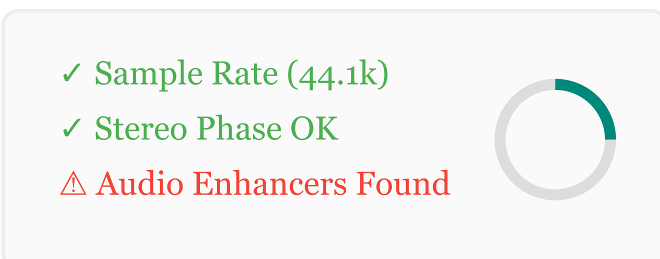



*Figure: The validation audit checks for common OS-level audio errors.*

- **Notch Depth Check:** Verifies your filters are achieving at least 40dB of attenuation.
- **Phase Test:** Ensures L/R channels are not inverted.
- **Sample Rate Check:** Confirms your system is not running in low-fidelity "Hands-Free" mode.

# 23. AUDIO WATCHDOG

**Automated Audio Watchdog:** Browsers often "suspend" audio in background tabs to save power. This background observer detects stalls and automatically attempts to resume the session, ensuring uninterrupted therapy even when the screen is off.

## Audio Status Indicator

Located in the top navigation bar, this provides real-time engine feedback:

- ● **Audio Active:** The engine is processing sound normally.
- ● **Audio Suspended:** The browser has paused the audio. Simply interact with the page (click any button) to resume playback.
- **Recovery Notification:** If the engine detects a stall (silence during active therapy), a floating orange notification will appear at the top of the screen as the Watchdog attempts an automated restart.

# 24. CLINICAL FAQ

## Clinical FAQ

**Is this a medical device?** No. It is an educational and research tool. It is not intended to diagnose or treat medical conditions.
**How loud should the therapy be?** Always use the **Mixing Point**. Total masking hides the signal from the brain, preventing it from learning to ignore the tinnitus. Maintain levels **below 70dB** to avoid acoustic fatigue.
**Bluetooth Warning:** While convenient, Bluetooth introduces **latency (delay)**. This is acceptable for simple noise, but can disrupt the precise 1.5Hz timing of **Acoustic CR** and the synchronization of **Bimodal Haptics**. Wired headphones are recommended for neuromodulation.
**Sample Rate Check:** Ensure your OS is set to 44.1kHz. Higher sample rates (96kHz+) are supported but unnecessary; rates below 44.1kHz (Bluetooth AG mode) will cause digital aliasing and render therapy ineffective.

## Stereo Verification Protocol

Precise stereo separation is required for **Decorrelation** and **Binaural Beats**. To verify your hardware:

1. Open the **System Validation** engine.
2. Run the **Phase Test**.
3. If the "In-Phase" tone sounds diffuse or wide, your system is incorrectly down-mixing to Mono or using "Virtual Surround." Disable these in your sound settings.

# 25. CLINICAL RESOURCES & MONITORING

The suite is built to facilitate seamless communication between patients and healthcare providers. Remote Patient Monitoring (RPM) can be configured to support clinicians using CPT code 99454.

## 25. CLINICAL RESOURCES & MONITORING

### How to Setup Email Reporting

1. **Open Settings:** On the main Dashboard, scroll to the bottom and click the gear icon to open **System Settings**.
2. **Enter Clinician Email:** Locate the field labeled **Audiologist Email**. Enter the professional email address provided by your clinician.
3. **Save Settings:** Click the **Save** button. The suite will now authorize a new "Email to Clinician" button on your dashboard.
4. **Dispatching a Report:** When it is time for your check-in, click **Email to Clinician**. This will automatically open your default email app with your adherence and distress metrics pre-formatted for your doctor.

### White-Label Customization

Clinics can customize the suite by adding their own branding in the Settings menu. By entering a **Clinic Name** and uploading a **Logo**, all generated PDF reports will feature a professional clinical letterhead.

## 26. GLOSSARY OF CLINICAL TERMS

**Habituation:** The neuroplastic process where the brain filters out the tinnitus signal from conscious awareness.

**Lateral Inhibition:** A physiological process where stimulated neurons suppress the activity of their overactive neighbors (the mechanism behind Notch Therapy).

**Minimum Masking Level (MML):** The lowest volume level of noise required to just barely hide the tinnitus sound.

**Residual Inhibition (RI):** The temporary reduction or suppression of tinnitus volume following sound exposure.

**Q-factor:** A measure of the sharpness of an auditory filter; high Q indicates very tonal tinnitus.

**Recruitment:** A condition where loudness perception increases abnormally fast, often associated with hearing loss and hyperacusis.

# 27. REPORTS & INTERPRETATION GUIDE

The suite provides a variety of reporting tools to help you and your clinician track progress. Understanding these reports is critical for effective self-management and professional clinical review.

## Catalog of Available Reports

### CHART LEGEND & KEY

**Primary Trend:** THI, RI, Q-Factor **Secondary Trend:** MML, Adherence, LG (Improved)

**Baseline:** Initial readings or population averages

- **Clinical Progress Report (PDF):** The primary document for professional review. It features high-resolution vector charts and a clinical summary. *(Generate on the main Dashboard)*
- **Clinical Email Dispatch:** A text-formatted version of your progress data sent via your device's email client. Ideal for Remote Patient Monitoring (RPM) check-ins. *(Configure your clinician's email in System Settings first)*
- **Module-Specific Exports (.txt):** Specialized text files containing raw data for tests like the TMC or RI. Use these when your doctor needs precise psychophysical values. *(Found on individual therapy pages)*
- **Personalized AI Insights:** A narrative report that correlates your distress scores with your usage logs and sleep reports to identify trends. *(Found in the AI Insights module)*
- **Q-Factor Trend Chart:** A specialized chart tracking the "sharpness" of your tinnitus signal over time. *(Found in the Progress History/Stats page)*
- **System Validation Report:** A technical audit confirming your hardware and browser are functioning correctly for therapy. *(Found in System Validation)*
- **Community Setup Summary:** A formatted summary of your therapy settings designed for sharing on forums like Tinnitus Talk. *(Click 'Share Setup' in any module)*

## How to Send Reports to Your Clinician

Communication with your healthcare provider is automated through two primary channels:

# 27. REPORTS & INTERPRETATION GUIDE

1. **One-Click Email:** If configured in **System Settings**, a button will appear on your dashboard to dispatch a text-based report instantly.
2. **PDF Submission:** Generate a **Clinical Progress Report (PDF)** to print for physical appointments or attach to a secure patient portal message.

## Understanding Your Progress Metrics

- **The Executive Clinical Summary:** An AI-synthesized paragraph highlighting statistically significant changes in your condition, saving your doctor time during review.
- **THI Score (Tinnitus Handicap Inventory):**
  Measures the impact of tinnitus. A decrease of **7 points or more** is the clinical marker for significant improvement.
  **Severity Bands:** 0-16 (Slight), 18-36 (Mild), 38-56 (Moderate), 58-76 (Severe), 78-100 (Catastrophic).

EXAMPLE: DISTRESS TREND (THI HISTORY)

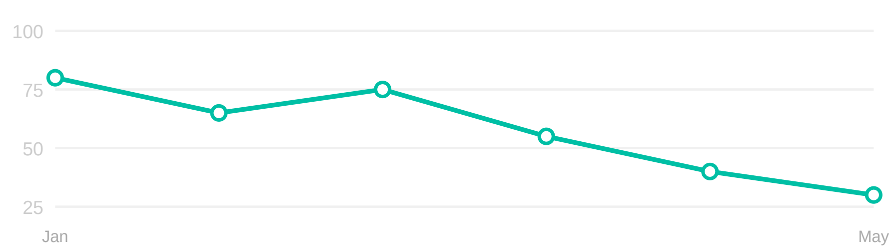



***Figure 1:*** *THI Distress Trend Chart showing positive habituation progress.*

- **MML (Minimum Masking Level):**
  **Interpretation:** The lowest volume required to hide your tinnitus. As you habituate, your brain learns to ignore the signal, causing the MML to drop over time. A lower MML means your tinnitus has less "contrast" against background sound.

# 27. REPORTS & INTERPRETATION GUIDE

**EXAMPLE: MINIMUM MASKING LEVEL (MML) TREND**

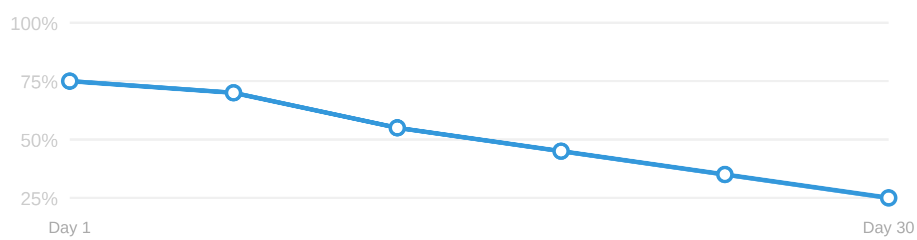



***Figure 2:*** *Minimum Masking Level (MML) Trend. Decreasing levels reflect clinical progress.*

- **Q-Factor Trend (Tonality Monitoring):**
  **Interpretation:** Tracks the "sharpness" of your tinnitus signal over time. Shifts in Q-factor provide objective evidence of cortical reorganization.
  - **Downward Trend:** Indicates the tinnitus is becoming more noise-like or "diffuse." This is a positive sign of habituation, as the neural representation of the sound is broadening.
  - **Upward Trend:** Often observed during the initial calibration phase. An increasing Q-factor suggests you are achieving more precise **Pitch Matching**, which allows for more surgical energy removal in Notch therapy.

**EXAMPLE: Q-FACTOR PROFILE SHIFT**

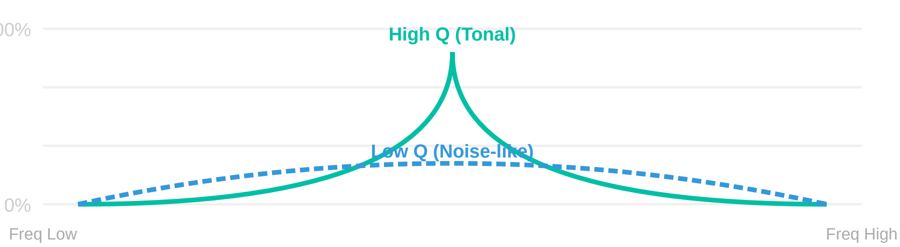



***Figure 3:*** *Q-Factor Profile Shift. Monitoring these changes on the Stats page helps track the "broadening" of the tinnitus signal as you habituate.*

- **RI Duration (Suppression):**

## 27. REPORTS & INTERPRETATION GUIDE

**Interpretation:** How long your tinnitus remains quiet after sound exposure. Increasing RI duration is a strong sign of positive neural plasticity.

EXAMPLE: SUPPRESSION DURATION (RI) TREND

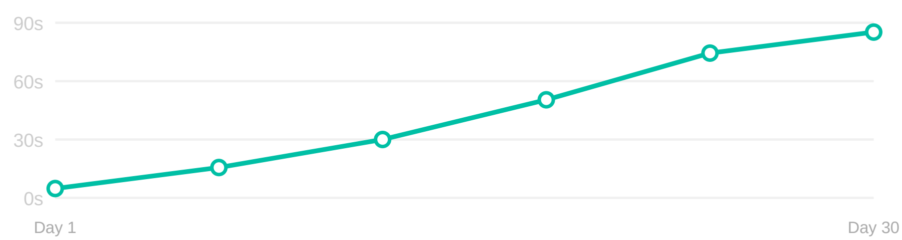



***Figure 4:*** *Residual Inhibition (RI) Suppression Trend indicating neural plasticity.*

- **LG (Loudness Growth):**
  **Interpretation:** Measures subjective loudness relative to objective volume. A "flattening" of this curve indicates a reduction in hyperacusis or sound sensitivity.

EXAMPLE: LOUDNESS GROWTH (LG) SHIFT


***Figure 5:*** *Loudness Growth (LG) Shift demonstrating improved sound tolerance.*

- **Adherence Trend:**
  **Interpretation:** Tracking daily therapy minutes. Consistency (30-60 minutes daily) is the strongest predictor of habituation success. Gaps in usage often correlate with temporary spikes in distress.

# 27. REPORTS & INTERPRETATION GUIDE

EXAMPLE: ADHERENCE TREND (LAST 30 DAYS)

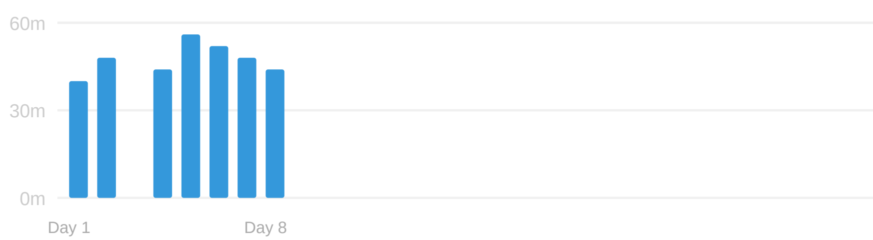



***Figure 6:*** *30-Day Adherence Bar Chart tracking daily therapy minutes.*

# 28. USER EXPERIENCE TIPS

## Glossary of UI Icons

- ● **Audio Active:** The audio engine is running normally and producing sound.
- ● **Audio Suspended:** The browser has paused the audio. Click anywhere on the page to resume.
- ○ **Audio Closed:** The engine is offline.
- ✓ **Verified:** The setting or hardware test is correctly calibrated.
- 🌓 **(Theme Toggle):** Switches between Light and Dark mode.
- ⚙ **(Gear Icon):** Opens the System Settings and Backup menu.

## Interactive Tutorials & Professional Narrator

Every module in the suite features an interactive walkthrough system to ensure you are calibrated correctly.

- **Narrated Guidance:** High-quality neural voice synthesis provides audio instructions for each step.

# 28. USER EXPERIENCE TIPS

- **Auto-Play Mode:** When enabled, the tutorial will automatically speak the instructions and advance through the setup sequence.
- **Visual Highlighting:** The tutorial will dim the screen and highlight specific sliders or buttons as they are discussed.
- **Narrator Controls:** You can toggle the narrator and adjust its volume or speed (0.5x to 2.0x) directly within the tutorial card.

## Layout & Themes

- **Compact Mode:** Reduces font and button sizes for users who prefer a high-density, technical interface. (Toggle in Settings).
- **Dashboard Layout:** Switch between a 1-column (mobile-optimized) or 2-column (desktop) view of your therapy cards.
- **Light/Dark Mode:** The suite defaults to a medical dark mode to reduce eye strain, but a high-contrast light mode is available.

## Common Mistakes to Avoid

- **Mistake 1: Over-Masking.** Setting the volume so you can no longer hear your tinnitus. *Correction:* Always find the **Mixing Point** so your brain can learn the signal is neutral.
- **Mistake 2: Bluetooth "Hands-Free" Mode.** Many headsets switch to a low-quality (8kHz) mono mode during calls. *Correction:* Ensure your OS output is set to **Stereo** (44.1kHz or 48kHz) to prevent filter distortion.
- **Mistake 3: Octave Confusion.** Matching your tinnitus pitch to a harmonic (usually double the actual frequency). *Correction:* Use the **x2** and **/2** buttons in the Notch Finder; the lowest clear match is usually the fundamental.
- **Mistake 4: OS Audio Enhancers.** Using "Bass Boost" or "Spatial Sound" (Dolby Atmos/Windows Sonic). *Correction:* Disable these settings; they alter the precise frequencies required for Notch and CR therapy.

# 28. USER EXPERIENCE TIPS

## Troubleshooting Quick Reference

| If you experience... | Recommended Action |
|---|---|
| **Audio cutting out on mobile** | Install as a PWA (Home Screen) and disable "Battery Optimization" for your browser. |
| **Sound feeling "irritating"** | Lower volume to the **Mixing Point** or switch to **Brown Noise** or **Rain**. |
| **Louder tinnitus after a session** | This is an "Autonomic Rebound." Use **Sleep Fade** and ensure therapy volume isn't too high. |
| **Buttons being unresponsive** | Check the **Audio Status Indicator**. If "Suspended," click the page to resume. |

## Recommended Daily Routine

🌅 **Morning (Reset):** 5 mins of 4-7-8 breathing + log your mood in the **CBT & Wellness** module.

💻 **Day (Passive):** Use **Broadband Sound** or **Decorrelated Noise** at a low level (background "wallpaper").

🌆 **Evening (Active):** 30-60 mins of **Notch** or **CR Therapy** while reading or relaxing.

🌙 **Bedtime (Sleep):** Use **Binaural Beats (Delta)** layered with **Ocean**. Enable **Sleep Fade**.

## Spike Survival Checklist

1. **De-escalate:** Use the **4-7-8 Breathing Pacer** for 2 minutes to calm your nervous system.
2. **Reframe:** Open **Cognitive Restructuring**. Log your reaction and use "AI Balanced Thought" to reduce the threat response.
3. **Enrich:** Do not sit in silence. Use low-level **Rain** or **Pink Noise**, keeping volume strictly below the tinnitus.

# 28. USER EXPERIENCE TIPS

4. **Patience:** Remind yourself that spikes are temporary neuroplastic events and usually subside within 48-72 hours.

## Managing Your Progress

**How to interpret your progress:**

- **The 7-Point Rule:** A drop of 7+ points in your monthly **THI Score** is the clinical marker of success.
- **Auditory Blind Spots:** Note times when you "forgot" your tinnitus for 15+ minutes while busy. This is the goal of habituation.
- **MML Reduction:** If the **Minimum Masking Level** trend is downward, your brain is successfully filtering the signal.

**When to reduce therapy:** If your THI score remains in the "Slight" range (0-16) for two consecutive months, you may begin tapering active sessions from daily to 2-3 times per week, using sound only as needed for comfort.

## Communicating with your Clinician

The Trahreg Suite is designed to support your relationship with your Audiologist or ENT. To get the most out of your appointments:

- **Bring the PDF:** Export and print your **Clinical Progress Report** every 30 days.
- **Discuss Trends:** Highlight changes in your THI scores and MML levels rather than daily volume fluctuations.
- **Mention Hearing:** If you notice shifts in your **Hearing Range Test**, your clinician may need to adjust your hearing aid settings or notch calibration.

**Home Screen Install:** Use "Add to Home Screen" on iOS/Android to install the suite as a PWA. This improves background audio reliability and removes the address bar.
**Focus Mode:** Enable "Do Not Disturb" during sessions. Incoming notifications can cause the volume to "duck" or drop, disrupting the calibrated Mixing Point.

# 29. HIPAA COMPLIANCE & PRIVACY

The Trahreg Suite is engineered with a "Zero-Knowledge" architecture to ensure maximum privacy and strict adherence to clinical data standards, including HIPAA (Health Insurance Portability and Accountability Act).

DIAGRAM: LOCAL-ONLY DATA FLOW


*Figure: Data is stored and processed entirely on your local hardware.*

### HIPAA Compliance Statement

Because the Trahreg Suite does not transmit, store, or process Protected Health Information (PHI) on centralized servers, it is inherently compliant with HIPAA regulations for clinical use. The application functions entirely within the patient's local browser environment. The developers of this suite have no access to patient identities, clinical scores, or therapy logs.

### 1. Zero-Knowledge Infrastructure

Standard medical apps store data in a cloud database. The Trahreg Suite does not. Your data lives exclusively in your browser's **Origin-Private Storage**. If you delete your browser data or use a different device, your logs are not "synced"—they remain only where you created them.

### 2. Clinical Anonymization (AI Features)

When utilizing the Google Gemini AI features for pattern analysis or clinical summaries, the suite performs a "Sanitization Pass":

- **PII Removal:** Names, emails, and IP addresses are never included in AI prompts.

## 29. HIPAA COMPLIANCE & PRIVACY

- **Metric-Only Transmission:** Only raw numeric values (THI: 45, Usage: 30m) are sent to identify trends.
- **Stateless Processing:** Data is processed as a "one-time" request and is not used to train the global model when using private API keys.

### 3. Patient-Controlled Encryption

If you choose to store your API key within the suite, it is protected via **AES-256 GCM encryption**. The decryption key is derived from your personal PIN using the **PBKDF2** standard (100,000 iterations) and is never stored on a server. This ensures that even if someone gained physical access to your device, your AI credentials remain protected.

### 4. Professional Reporting Security

The **Clinical Progress Report (PDF)** is generated locally using the `html2pdf` engine. The data does not "round-trip" to a server to be rendered into a document. This ensures that the sensitive relationship between you and your healthcare provider remains private.

### 5. AI Capabilities & Limitations

The Gemini AI integration serves as a therapeutic assistant, not a clinical diagnostic tool.

- **What it CAN do:** Reframe distressing thoughts using TRT principles, suggest sound recipes based on descriptions, and identify statistical correlations in your local history.
- **What it CANNOT do:** Diagnose hearing loss, prescribe medical treatments, or replace the expertise of an Audiologist or ENT.
- **Technical Note:** AI can occasionally "hallucinate" or provide inaccurate information. Always verify technical settings (like pitch) manually using the suite's diagnostic tools.

#### ⚠ Prominent Clinical Warning

**The AI Assistant is for educational exploration only. It is NOT a substitute for professional medical advice. If you experience sudden hearing loss, severe vertigo, or suicidal ideation, contact emergency services or your healthcare**

## 29. HIPAA COMPLIANCE & PRIVACY

**provider immediately.**

## 30. DATA PORTABILITY: BACKUP & RESTORE

Because the Trahreg Suite is a "Zero-Knowledge" platform, your data exists only on your specific device. If you switch computers or clear your browser history, your progress will be lost unless you have a backup.

### How to Create a Secure Backup

1. **Open Settings:** Navigate to the **System Settings** (gear icon) on the main dashboard.
2. **Export Data:** Find the **Backup & Restore** section and click **Download System Backup (JSON)**.
3. **Storage:** Your browser will download a file named `trahreg_tinnitus_backup_[DATE].json`. Because this file contains your clinical scores and therapy logs, store it in a secure location (e.g., an encrypted USB drive or a password-protected cloud folder).

### How to Restore Your Progress

1. **Locate Backup:** Ensure you have your `.json` backup file available.
2. **Import Data:** In **System Settings**, click **Restore from File** and select your backup.
3. **Refresh:** The suite will automatically reload, and all your history, THI scores, and therapy calibrations will be restored.

### Best Practices for Data Safety

- **Monthly Backups:** Create a backup after every THI assessment (every 30 days) to ensure you don't lose clinical trend data.

## 30. DATA PORTABILITY: BACKUP & RESTORE

- **Browser "Incognito" Warning:** Data is often cleared automatically when closing an Incognito/Private window. Always use a standard browser window for therapy to ensure persistence.
- **Avoid Shared Computers:** If using a public or shared computer, always use the **Full System Reset** tool in Settings after exporting your backup to remove your clinical footprint.

## 31. RESEARCH & SCIENTIFIC PRINCIPLES

The suite includes a dedicated **Research** page that summarizes the clinical evidence and physiological theories behind each implemented modality. This includes detailed explanations of:

- **Lateral Inhibition:** The mechanism for Notch Therapy.
- **Neural Desynchronization:** The goal of Acoustic CR and Decorrelated noise.
- **Multisensory Integration:** The principle behind Bimodal Neuromodulation.

Direct links to peer-reviewed studies (Okamoto, Tass, Conlon, etc.) are provided for clinicians and researchers to verify technical parameters.

## 32. HARDWARE & DEVELOPER TOOLS

**Hardware & Testing (Dev):** A restricted utility area for technical verification. This page contains tools for testing LRA haptic motor mounting, 3D model visualization for hardware assembly (Finger Pacer), and diagnostic signal validation loops.
*Note: This section is intended for developers and professional partners involved in the hardware prototyping phase.*

## 33. MILESTONE ACHIEVEMENT CERTIFICATE

**Purpose:** Reaching "Full Habituation" is a significant clinical milestone. To provide positive psychological reinforcement, the suite includes an automated celebration system.

- **The Celebration:** When you record a THI score in the "Slight" range (0-16) or manually check the final milestone in the Stats page, the application triggers a celebratory confetti burst.
- **The Certificate:** Reaching Phase 6 unlocks a downloadable **Achievement Certificate (PDF)**. This professional document can be accessed on the main Dashboard or via the **Doctor's Summary** page. It can be printed to mark your success or shared with your healthcare provider as part of your clinical history.

## APPENDIX A: TECHNICAL APPENDIX

### Flexible Audio Sources

Every therapy module supports three input methods:

- **Internal:** Calibrated white, pink, and brown noise.
- **Custom:** User-uploaded relaxing music or nature files.
- **Live:** Capture audio from other browser tabs (Spotify/YouTube) and process it through the Notch or Decorrelation engine in real-time.

### Noise Generator (Python-style)

A specialized utility for generating broadband noise with scientific precision, mirroring Python signal-processing behavior. Uses Paul Kellet filter banks for Pink noise and lossy integrators for Brown noise.

# APPENDIX A: TECHNICAL APPENDIX

## Table 1: Protocol Quick Start Summary

| Modality | Key Setting | Target Volume | Duration |
|---|---|---|---|
| **Notch Therapy** | Match Freq / 1.0 Octave | Mixing Point | 30-60m Daily |
| **Acoustic CR** | Log-spaced Tones | Equal Loudness | 60m Daily |
| **Decorrelated Noise** | Pink Noise / Wide Stereo | Mixing Point | 24/7 Enrichment |
| **Dual-Stimulus** | Haptic Sync / Mobile | Comfortable | 30m Daily |
| **Broadband Sound** | Pink Noise / Nature | Mixing Point | 24/7 Enrichment |
| **Binaural Beats** | Delta/Theta Offset | Comfortable | Sleep Transitions |

# APPENDIX B: QUICK REFERENCE FOR CLINICIANS

This appendix is designed for audiologists and ENTs using the Trahreg Suite for patient monitoring and auditory rehabilitation.

## Clinical Metrics Thresholds

- **THI Significance:** A drop of **≥ 7 points** in the Tinnitus Handicap Inventory is the minimum clinically significant difference (MCSD).
- **Q-Factor Interpretation:** High Q values (> 3.0) suggest tonal tinnitus localized to specific neural clusters. Low Q values (< 1.5) suggest a broader, noise-like perception involving multiple auditory channels.
- **RI Depth:** Patients exhibiting "Total RI" (complete silence) typically show higher responsiveness to neuromodulation protocols (ACR/Notch).

## Therapy Selection Matrix

| Symptom Profile | Primary Recommendation |
|---|---|
| High Distress (THI > 58) | CBT & Wellness / Mindfulness |
| Tonal / High Q-Factor | Notch Therapy / Acoustic CR |
| Noise-like / Low Q-Factor | Decorrelated Noise / TRT Masking |
| Sleep Disturbance | Binaural Beats (Delta/Theta) |
| Hyperacusis (Steep LG) | Passive Enrichment / Desensitization |

## Remote Monitoring (CPT 99454)

Clinicians can utilize the suite's automated reporting to satisfy requirements for **CPT 99454** (Remote monitoring of physiologic parameter(s) via patient-managed device). Documentation provided in the Clinical PDF includes:

- Patient adherence logs (daily therapy minutes).

# APPENDIX B: QUICK REFERENCE FOR CLINICIANS

- Monthly distress assessments (THI).
- Serial psychophysical measures (MML/RI).

*Clinicians should verify specific documentation requirements with their local payers.*

### Protocol Compliance Checklist (Patient)

Use this checklist to ensure adherence to clinical habituation protocols. Consistency is the primary driver of successful neuroplasticity.

| Requirement | Frequency | Verified |
| --- | --- | --- |
| **Therapy Adherence:** 30-60 mins of daily sound therapy sessions. | Daily | [ ] |
| **Mixing Point:** Volume is set so noise and tinnitus "mix" (not masked). | Daily | [ ] |
| **System Integrity:** Performed weekly System Validation and Phase audit. | Weekly | [ ] |
| **Handicap Tracking:** Logged monthly THI score to track progress. | Monthly | [ ] |
| **Data Portability:** Generated and stored a secure JSON backup file. | Monthly | [ ] |

# APPENDIX C: TROUBLESHOOTING GUIDE

Most technical issues in sound therapy arise from OS-level audio processing or hardware limitations. Follow this guide to ensure your system is providing clinically accurate output.

### 1. Bluetooth Connectivity Issues

- **The "Hands-Free" Trap:** Many Bluetooth headphones have two modes: "Stereo" and "Hands-Free AG." **Stereo** provides high-fidelity audio (44.1kHz+). **Hands-Free** drops the quality to 8kHz, which destroys the therapy filters. Always ensure your OS output is set to the Stereo profile.
- **Latency:** Bluetooth has inherent delay. If using **Bimodal Haptics** (Lenire module), use wired headphones for the most precise synchronization between sound and vibration.

### 2. Audio "Ducking" & Volume Fluctuations

- **Notifications:** When your phone or PC receives a notification, it often "ducks" (lowers) the volume of other apps. This disrupts the **Mixing Point**. Enable **Do Not Disturb** or **Focus Mode** during therapy sessions.
- **Normalization:** Features like "Loudness Equalization" or "Sound Normalization" in Windows/macOS attempt to make all sounds the same volume. This will fight against the suite's calibrated output. Disable these in your System Sound settings.

### 3. Hardware Phase & Stereo Image

- **Centered Image:** If the **In-Phase** tone in the System Validation tool sounds "wide" or "hollow" instead of centered, your headphones may be wired incorrectly or you are using **Virtual Surround Sound**.
- **Action:** Disable 7.1 Surround, Dolby Atmos, or Windows Sonic while using the suite. These features introduce phase shifts that invalidate Decorrelation and Binaural therapies.

### 4. Browser Permission Blocks

- **Autoplay:** Browsers prevent sites from playing audio automatically. If you don't hear sound, click anywhere on the page to "resume" the audio context.

## APPENDIX C: TROUBLESHOOTING GUIDE

- **Tab Throttling:** If audio stops when you switch tabs, ensure you have the suite installed as a **PWA** (Add to Home Screen) or keep the tab visible. The built-in **Audio Watchdog** attempts to recover from these stalls automatically.

# APPENDIX D: PATIENT INTAKE FORM

Clinicians may use this form to establish a baseline for new patients entering an auditory habituation program. Patients should complete this prior to their initial consultation.

### Patient Information

Name: ________________________________________________ Date: _______________

Date of Birth: _________________________ Primary Care Physician: __________________

## Tinnitus Profile

**Perceived Location:** [ ] Left Ear [ ] Right Ear [ ] Both Ears [ ] Inside Head

**Onset:** When did you first notice the sound? ______________________________________

**Character:** [ ] High-pitched Ring [ ] Hiss [ ] Buzz [ ] Roar [ ] Pulsatile (matches heartbeat)

**Fluctuation:** [ ] Constant [ ] Intermittent [ ] Volume changes during the day

## Hearing & Medical History

**Hearing Loss:** Do you have difficulty understanding speech? [ ] Yes [ ] No

**Hearing Aids:** Do you currently use hearing aids? [ ] Yes [ ] No

**Sound Sensitivity:** Do everyday sounds (e.g., dishes, sirens) cause pain? [ ] Yes [ ] No

**Other Symptoms:** [ ] Vertigo/Dizziness [ ] Fullness in Ears [ ] Jaw/Neck Pain

## Current Impact

On a scale of 0-10 (10 being most severe), how much does tinnitus affect your:

## APPENDIX D: PATIENT INTAKE FORM

Sleep: ______ Concentration: ______ Quiet Relaxation: ______ Work/Social: ______

*Clinician Note: Attach the patient's initial THI (Tinnitus Handicap Inventory) score from the suite's assessment module to this form for a complete baseline.*

# APPENDIX E: HABITUATION MILESTONE TRACKER

Habituation is a gradual process. Use this tracker to identify your progress through the typical stages of auditory retraining. Note that progress is rarely linear; "spikes" are a normal part of the journey.

| Milestone Phase | Description of Experience | Target Date |
|---|---|---|
| **1. Baseline** | Initial THI score logged. Tinnitus is perceived as a constant threat or source of high distress. | ___________ |
| **2. Technical Mastery** | Consistently using therapy at the **Mixing Point**. Precise pitch match found via Notch Finder. | ___________ |
| **3. Psychological Shift** | Using CBT tools to reframe reactions. Reduced "fight or flight" response when noticing the sound. | ___________ |
| **4. Early Habituation** | First instance of "forgetting" the sound while busy. THI score shows a clinically significant drop. | ___________ |
| **5. Partial Habituation** | Sound is perceived as neutral background (fridge hum). Tinnitus is audible but no longer distressing. | ___________ |
| **6. Full Habituation** | Long periods (days/weeks) of complete unawareness. Quality of life has returned to baseline. | ___________ |

**Achievement Reward:** Reaching Phase 6 (Full Habituation) unlocks a celebratory confetti burst and a downloadable **Achievement Certificate**.

## APPENDIX E: HABITUATION MILESTONE TRACKER

**Certificate Sharing Protocol:** Upon reaching full habituation, it is recommended to download your certificate and share it with your healthcare provider. This professional record validates your progress and should be appended to your clinical history during follow-up reviews.

**Pro-Tip:** Identifying these milestones with your healthcare provider helps adjust your protocol for the next stage of recovery.

# APPENDIX F: EXPORT & REPORTING SAMPLES

Below are mockups of the various clinical exports available in the suite. These samples help you understand how your data is presented to healthcare providers.

## 1. Clinical PDF Report Mockup

### Clinical Progress Report

Trahreg Tinnitus Suite v2026.05.3

**Date:** May 30, 2026
**Patient ID:** [LOCALLY_STORED_ID]

**EXECUTIVE SUMMARY**

*"Patient reports a 12-point decrease in THI over 30 days. Adherence remains high (avg 45m/day). Suggest continuing Notch Therapy at 8400Hz..."*

[THI TREND CHART]

[MML TREND CHART]

## 2. Email-to-Clinician Sample

**Subject:** Tinnitus Clinical Report - [Your Name]

```
TRAHREG TINNITUS THERAPY SUITE - CLINICAL REPORT
App Version: 2026.05.3
Export Date: 5/30/2026, 1:32 AM
-------------------------------------------
THERAPY SETTINGS:
Mode: Notch Therapy
Center Freq: 8400 Hz
Volume: 35% (Mixing Point)

PSYCHOLOGICAL BASELINE:
```

# APPENDIX F: EXPORT & REPORTING SAMPLES

```
Last THI Score: 32/100 (Mild)
Change: -8 points (Clinically Significant)
-------------------------------------------
```

### 3. Raw Data Export (.txt)

```
# TMC Raw Data Export
# Freq(Hz), Level(%)
1000, 20
2000, 18
4000, 15
8000, 45
8400, 52 (Peak/Tinnitus)
10000, 22
12000, 15
```

# APPENDIX G: THERAPY MODALITY COMPARISON

| Therapy | Primary Target | Mechanism | Calib. Required |
|---|---|---|---|
| **Notch** | Tonal Tinnitus | Lateral Inhibition | High (Pitch Match) |
| **Acoustic CR** | Neural Synchrony | Desynchronization | Very High (Bracketing) |
| **Broadband** | Distress/Contrast | Habituation (TRT) | Low (Volume only) |
| **Decorrelated** | Spatial Image | L/R Independence | Medium (Stereo) |

## APPENDIX H: GLOSSARY OF SOUND ENGINEERING TERMS

- **Pink Noise:** A sound spectrum where every octave contains equal energy. Preferred for therapy as it mirrors natural acoustic distributions.
- **Brown Noise:** Higher energy at lower frequencies (bass heavy). Often perceived as "warmer" or "softer" than White noise.
- **Carrier Frequency:** The main "test tone" or audio signal that is being processed or modulated by another signal.
- **Amplitude Modulation (AM):** The systematic variation of volume over time, used in bimodal therapy to increase signal saliency.
- **LFO (Low Frequency Oscillator):** A slow-moving signal (0.1Hz - 20Hz) used to control parameters like volume or filter sweeps.
- **Logarithmic Scale:** A non-linear scale where equal distances represent equal ratios (e.g., octaves), matching how the human ear perceives pitch.

# APPENDIX I: RECOMMENDED HARDWARE & SETUP

Precise neuromodulation requires linear frequency response and high signal-to-noise ratios. Below are recommended hardware standards:

### 1. Headphones

- **Preferred:** Wired, Over-Ear, Open-Back (e.g., Sennheiser HD600 series, Beyerdynamic DT880/990). Open-back models provide superior spatial imaging for Decorrelation.
- **Secondary:** High-quality wired IEMs (In-Ear Monitors).
- **Avoid:** Noise-canceling headphones (ANC) during *calibration*, as their internal DSP can alter the phase of the therapy signal.

### 2. Audio Interfaces (DACs)

- Standard laptop/smartphone jacks are usually sufficient. For desktop users, a simple external USB DAC (like the Apple USB-C Dongle) provides a cleaner floor for high-frequency testing.

### 3. Mobile Considerations

- Use wired connections where possible to eliminate the 150ms-300ms latency inherent in Bluetooth, which can desynchronize Bimodal Haptics.

## APPENDIX J: DIAGNOSTIC TEST INTERPRETATION GUIDE

| Diagnostic Test | What a DOWNWARD trend means | Clinical Significance |
|---|---|---|
| **THI Score** | Reduced psychological distress. | Success (Habituation) |
| **MML (Volume)** | Lower subjective tinnitus volume. | Success (Auditory Gain) |
| **LG (Loudness)** | Improved sound tolerance. | Reduced Hyperacusis |
| **Q-Factor Profile Shift** | Downward Trend: Signal is broadening (less tonal). | Positive Neural Change |
| **Hearing Profile (Audiogram)** | Improving sensitivity (values moving toward 0dB). | Reduced Auditory Deprivation |
| **Diagnostic Test** | **What an UPWARD trend means** | **Clinical Significance** |
| **RI Duration** | Longer suppression after sound. | Positive Plasticity |
| **Q-Factor Profile Shift** | Upward Trend: Increasingly tonal/narrow sound. | Better Notch Targeting |
| **Hearing Profile (Audiogram)** | Worsening loss (values moving toward 110dB). | Increased Deprivation / Shift in Therapy Target |

Access the suite at: **https://tinnitus.trahreg.com**